

Note: The difference between the reject rule and drop rule is that the former will send back an ICMP packet detailing why a connection was rejected. A drop rule just drops the packet and does nothing else. Normally, an administrator will want to use a reject rule for friendly and neutral networks, and a drop rule only for hostile networks.

- `Firewall-cmd --permanent --zone=vnc --add-rich-rule='rulefamily=ipv4 source address=192.168.1.0/24 port port=7900-7905 protocol=tcp accept'`

This accepts all TCP packets on Ports 7900, up to and including Port 7905, in the VNC zone for the 192.168.1.0/24 subnet.

Logging with rich rules

When debugging or monitoring a firewall, it can be useful to have a log of accepted or rejected connections. FirewallD can accomplish this in two ways — by logging to *syslog*, or by sending messages to the kernel audit sub-system, managed by *auditd*.

In both cases, logging can be rate limited, which ensures that system log files do not fill up with messages at a rate at which the system cannot keep up or fills all its disk space.

The basic syntax for logging to *syslog* using rich rules is:

```
Log [prefix="PREFIXTEXT"] [level=<LOGLEVEL>] [limit
value="<RATE/DURATION>"]
```

...where *<LOGLEVEL>* is one of *emerg*, *alert*, *crit*, *error*, *warning*, *notice*, *info*, or *debugging*.

<DURATION> can be of 's' for seconds, 'm' for minutes, 'h' for hours, or 'd' for days. For example, the *limit value=3/m* will limit the log messages to a maximum of three per minute.

The basic system for logging the audit sub-system is:

```
Audit [limit value="<RATE/DURATION>"]
```

The rate limiting is configured in the same way as for *syslog* logging.

Logging examples

Some examples of logging using rich rules are shown below:

```
# firewall-cmd --permanent --zone=work --add-rich-rule='rule
servicename="ssh" log prefix ="ssh" level="notice" limit value
="3/m" accept
```

The above code indicates that you can accept new connections to ssh from the work zone, and log new connections to *syslog* at the notice level with a maximum of three messages per minute.

```
#firewall-cmd --add-rich-rule='rule family=ipv6 source
address="2001:db8::/64" service name="dns" audit limit
value="1/h" reject' --timeout=300
```

The code given above states that new IPv6 connections from the subnet 2001:db8::/64 in the default zone to DNS are rejected for the next five minutes, and rejected connections are logged to the audit system with a maximum of one message per hour. **END** 

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